Mammalia Raccoon Proximity Network Analysis

https://github.com/aiyshwary/Raccoon-Network-Analysis

Python NetworkX Jupyter Graph Analysis Epidemiology

OVERVIEW

A graph-based epidemiological study modelling rabies transmission risk in urban raccoon populations through proximity network analysis and centrality measures, supporting public health and wildlife management decisions.

IMPACT

Constructed and analyzed a raccoon proximity network using real-world contact data, applying centrality measures to identify high-risk individuals and quantify potential rabies spread pathways within urban populations.

TECHNICAL WALKTHROUGH

Mammalia Raccoon Proximity Network Analysis – Project Explanation Project Overview This project analyzes social interaction patterns among urban raccoons using network analysis to understand how rabies and other pathogens can spread within raccoon populations.

Since raccoons are a major vector of rabies, studying how often and how closely they interact helps estimate disease transmission risk and supports better public health and wildlife management strategies.

Objective Model raccoon interactions as a graph/network Identify high-risk raccoons that can spread pathogens quickly Use centrality measures to understand social structure and disease flow Dataset & Network Construction 24 raccoons → represented as nodes ~2000 interactions → represented as weighted edges Data collected using proximity-detecting collars Edge weight = time spent in close proximity Monthly interaction data used to build the network

If two raccoons are connected, it means they had close physical

interaction, which is critical for rabies transmission.

Why Network Analysis?

Disease spread is not random.

It depends on: Who interacts with whom How frequently Who acts as a bridge between groups Network analysis helps identify super-spreaders and critical connectors.

Centrality Measures Used & Interpretation

Degree Centrality

Measures number of direct connections High degree = more interactions Inference: Raccoon #4 has the highest degree Acts as a high-contact individual High risk of initiating widespread infection

Closeness Centrality

Measures how close a node is to all others Indicates how fast something can spread Inference: Raccoon #2 has highest closeness centrality Can spread pathogens rapidly across the entire network

Betweenness Centrality

Measures how often a node lies on shortest paths Identifies bridge nodes Inference: Raccoon #16 connects different subgroups Acts as a key transmission bridge Removing or vaccinating this raccoon can significantly slow disease spread

Clustering Coefficient

Measures how tightly a raccoon is embedded in a local group Indicates family or group behavior Inference: Raccoon #2 has highest clustering Infection here may cause localized outbreaks within families

Eigenvector Centrality

Measures influence based on connections to influential nodes Inference: Raccoon #4 again ranks highest Connected to other highly interactive raccoons Considered a super-spreader candidate

PageRank Algorithm

Evaluates overall importance considering direction and weight of interactions Helps rank raccoons by global influence Key Findings Raccoon populations are highly interconnected Disease can spread much faster than expected A few raccoons play disproportionately large roles in pathogen transmission Targeted intervention (vaccination/tracking of key raccoons) is more effective than random control Technologies & Concepts Used Graph Theory NetworkX (Python) Weighted graphs Centrality algorithms Epidemiological modeling concepts

KEY DETAILS

- Built a weighted proximity network from the mammalia-raccoon-proximity dataset using Python and NetworkX.
- Computed degree, betweenness, closeness, and eigenvector centrality to rank individuals by transmission risk.
- Identified superspreader nodes whose removal would most disrupt disease propagation chains.
- Visualized network topology and centrality distributions to communicate findings clearly.
- Results provide evidence-based recommendations for targeted wildlife intervention strategies.